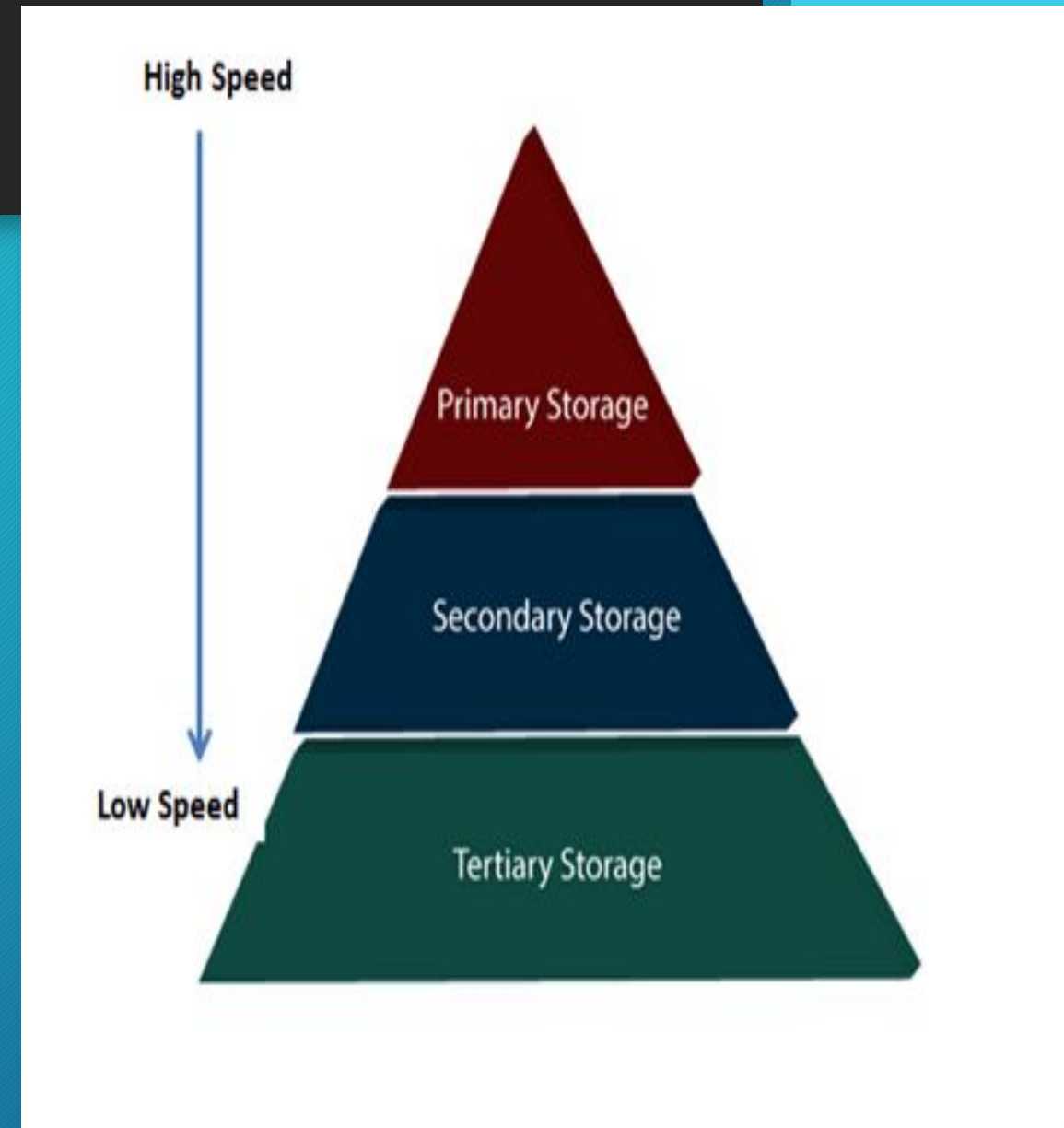


Store data in Disk and Files



Storage System in DBMS

- A database system provides an ultimate view of the stored data.
- However, data in the form of bits, bytes get stored in different storage devices.
- we will take an overview of various types of storage devices that are used for accessing and storing data.
- **Types of Data Storage**
- These storage types differ from one another as per the speed and accessibility.
- Primary Storage
- Secondary Storage
- Tertiary Storage



Primary Storage

- It is the primary area that offers quick access to the stored data. We also know the primary storage as volatile storage. Main memory and cache are the types of primary storage.
- **Main Memory:** It is the one that is responsible for operating the data that is available by the storage medium. The main memory handles each instruction of a computer machine.
- **Cache:** It is one of the costly storage media. On the other hand, it is the fastest one. A cache is a tiny storage media which is maintained by the computer hardware usually

Secondary Storage

- It is the storage area that allows the user to save and store data permanently. This type of memory does not lose the data due to any power failure or system crash. That's why we also call it non-volatile storage.
- There are some commonly described secondary storage media which are available in almost every type of computer system:
- **Flash Memory:** A flash memory stores data in USB (Universal Serial Bus) keys which are further plugged into the USB slots of a computer system. This type of memory storage is most commonly used in the server systems for caching the frequently used data. This leads the systems towards high performance and is capable of storing large amounts of databases than the main memory.
- **Magnetic Disk Storage:** A magnetic disk is used for storing the data for a long time. It is capable of storing an entire database. It is the responsibility of the computer system to make availability of the data from a disk to the main memory for further accessing.
- Also, if the system performs any operation over the data, the modified data should be written back to the disk. The tremendous capability of a magnetic disk is that it does not affect the data due to a system crash or failure.

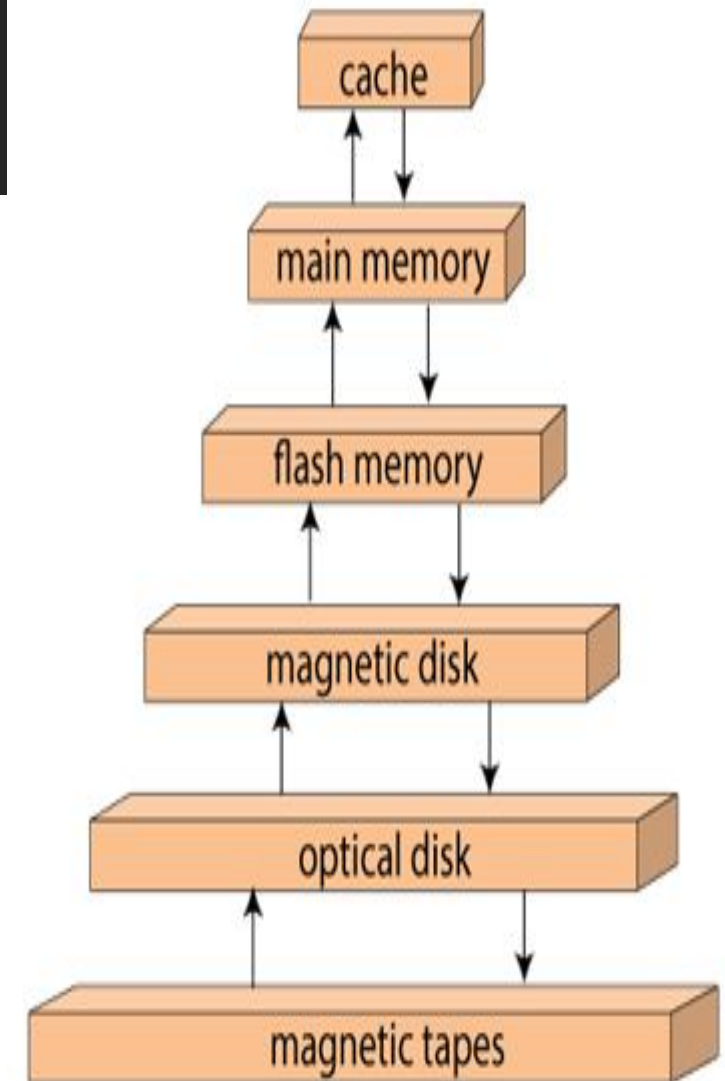
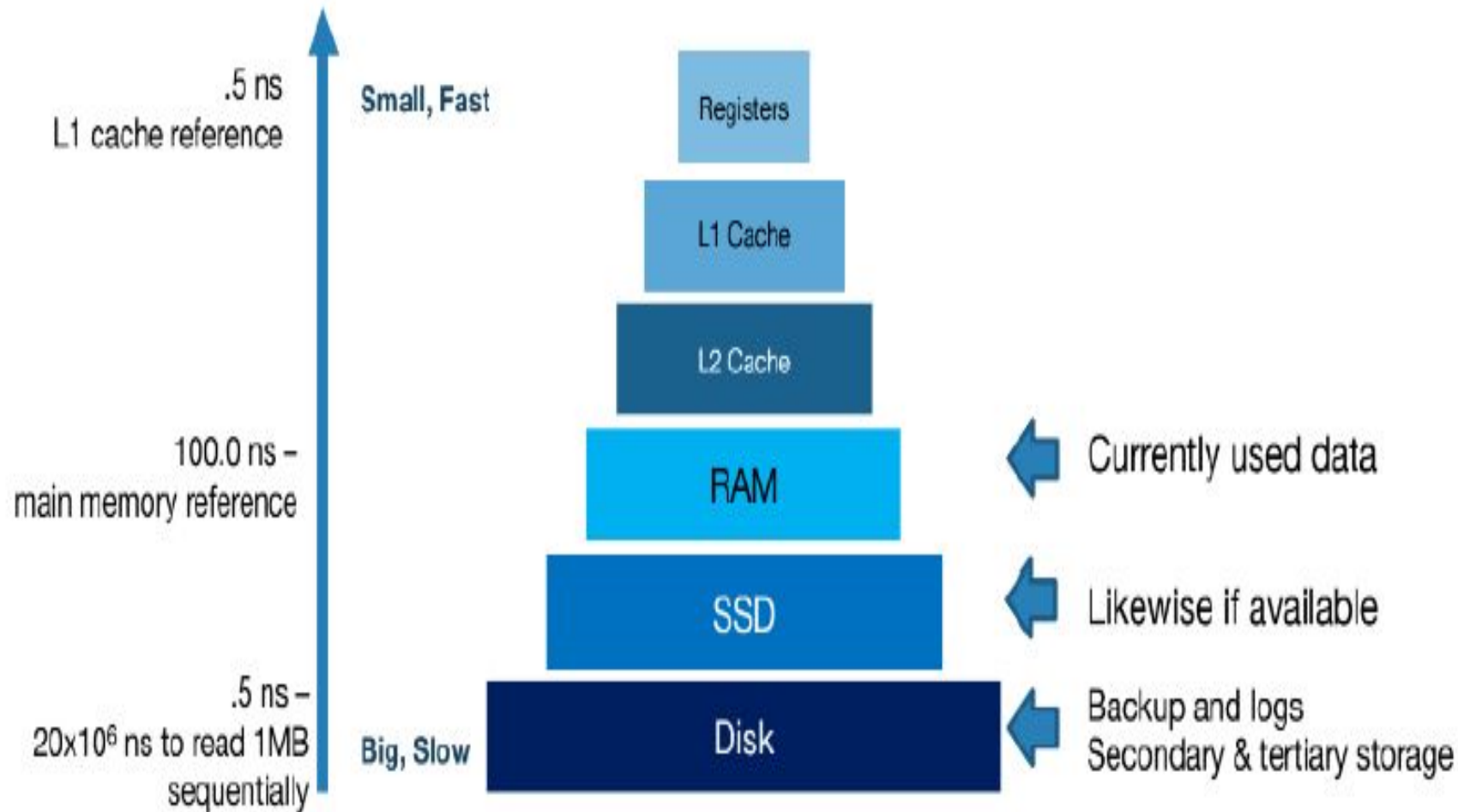
Memory and Disk

- Whenever a database uses data, that data must exist in memory. Accessing this data is relatively
- fast, but once the data becomes very large, it becomes impossible to fit all of it within memory.
- Disks are used to cheaply store all of a database's data, but they incur a large cost whenever data is accessed or new data is written.
- The basic API for disks includes READ and WRITE which stands for transferring “pages” of data
- from disk to RAM and transferring “pages” of data from RAM to disk respectively. Note that both
- API calls are very slow due to the structure of magnetic disks.

Tertiary Storage

- It is the storage type that is external from the computer system. It has the slowest speed. But it is capable of storing a large amount of data. It is also known as Offline storage. Tertiary storage is generally used for data backup. There are following tertiary storage devices available:
- **Optical Storage:** An optical storage can store megabytes or gigabytes of data. A Compact Disk (CD) can store 700 megabytes of data with a playtime of around 80 minutes. On the other hand, a Digital Video Disk or a DVD can store 4.7 or 8.5 gigabytes of data on each side of the disk.
- **Tape Storage:** It is the cheapest storage medium than disks. Generally, tapes are used for archiving or backing up the data. It provides slow access to data as it accesses data sequentially from the start. Thus, tape storage is also known as sequential-access storage. Disk storage is known as direct-access storage as we can directly access the data from any location on disk.

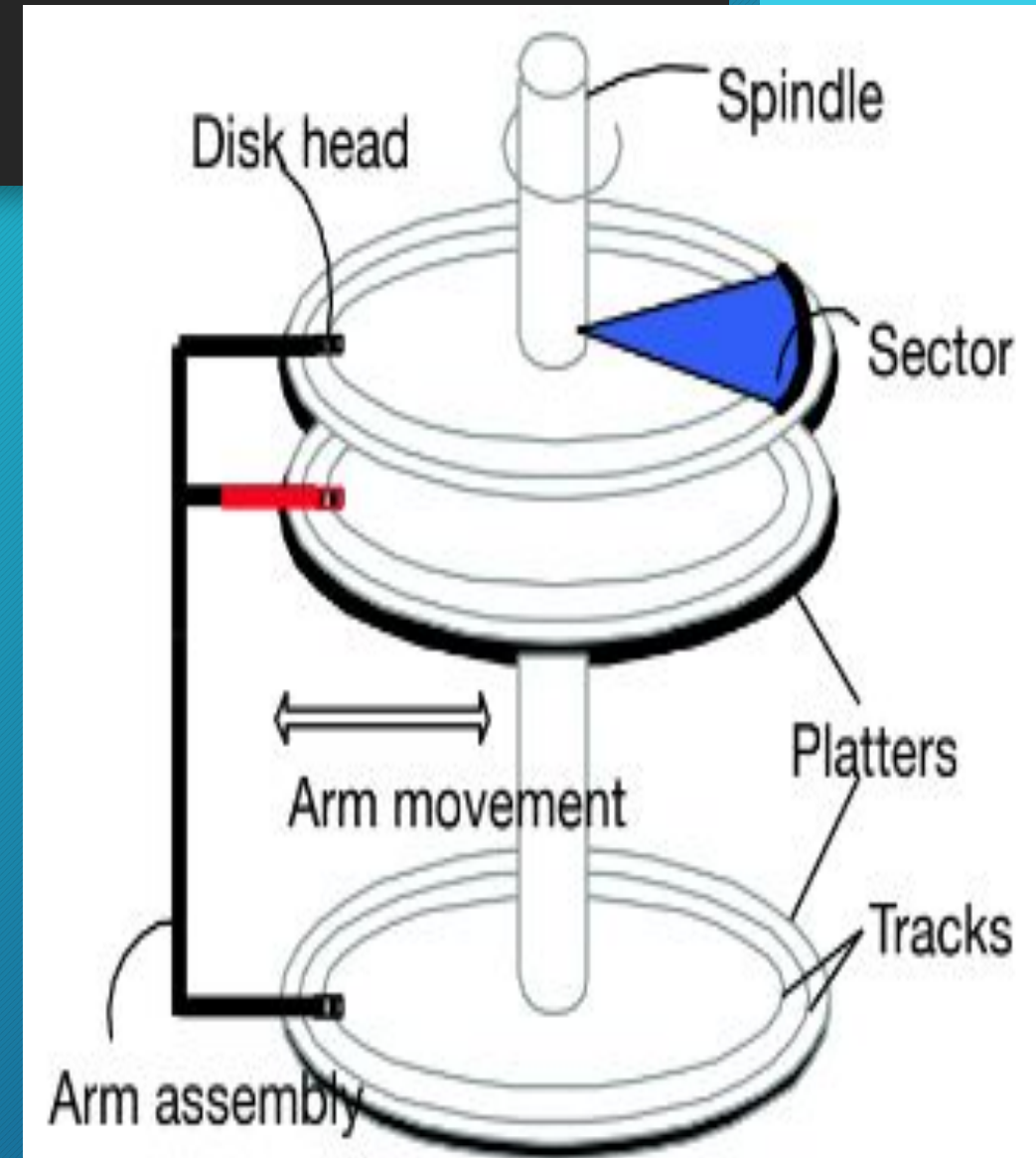
Storage Hierarchy



Storage device hierarchy

Disk Structure

- **Platters** usually spin at around 15000 rpm. The **arm assembly** moves in or out to position a head
- on a desired **track** which is under heads and makes a “cylinder”. Only one head reads/writes at any one time.
- The block/page size is a multiple of (fixed) **sector** size.



Accessing a Disk Page

- Times to access (read/write) a disk block are listed as below:
- seek time (moving arms to position disk head on track); 2-3 ms on average
- rotational delay (waiting for block to rotate under head); 0-4 ms (15000 RPM)
- transfer time (actually moving data to/from disk surface); 0.25 ms per 64KB page

Disks vs. SSD

- Solid State Drives (SSD), or Flash, is another medium to store data. Different from magnetic disks, SSDs are organized into cells, and supports fast random reads.
- Note that in contrast to SSDs, hard disks perform very bad on random reads because spatial locality is important for disks and sequential reads are much faster.
- SSDs are generally more expensive than hard disks too

Data Storage and Querying in DBMS

- Through DBMS, users can manage the data efficiently in a database to increase accessibility and productivity.
- So, a Database System is a software system, having an organized collection of structured information, stored in a computer system
- The Database System is further divided into two components. They are as follows:
 - Data Storage Manager
 - Query Manager

Data Storage Manager

- Data Storage Manager also known as “Database Control System”, is generally a program that provides an interface between the data/information stored and the queries received.
- Storage Manager is generally in charge of the interactions with the File Manager, where raw data is stored on the data with the help of the file system. It translates various DML statements into low-level commands.
- **File Manager:** The main purpose of File Manager is to manage the allocation of space on the disk storage.
- **Buffer Manager:** The main purpose of Buffer Manager is to fetch the data from disk storage into the main memory. The buffer manager is a critical part of the database system since it enables the database to handle data sizes that are much larger than the size of the main memory.

storage manager

The storage manager implements several data structures as part of the physical system implementation:

- Data files, which store the database itself.
- Data dictionary, which stores metadata about the structure of the database, in particular the schema of the database.
- Indices, which can provide fast access to data items. Like the index in this textbook, a database index provides pointers to those data items that hold a particular value.
- For example, we could use an index to find the instructor record with a particular ID, or all instructor records with a particular name.
- Hashing is an alternative to indexing that is faster in some but not all cases.

Query Processor

- Query Processor contains the following three components:
- **DDL Interpreter:** which interprets DDL statements and records the definitions in the data dictionary.
- **DML Compiler:** performs query optimization; that is, it picks the lowest cost evaluation plan from among the alternatives.
- **Query Evaluation Engine:** It executes the low-level language instructions, generated by DML Compiler.

Disk Space Management

- Disk Space Management is the lowest layer of DBMS. It is in charge of managing space on disk.
- Its main purposes include mapping pages to locations on disk, loading pages from disk to memory, and saving pages back to disk and ensuring writes.



Files, Pages, Records

- The basic unit of data for relational databases is a **record** (row). These records are organized into **relations** (tables) and can be modified, deleted, searched, or created in memory.
- The basic unit of data for disk is a **page**, the smallest unit of transfer from disk to memory and vice versa.
- In order to represent relational databases in a format compatible with disk, each relation is stored in its own file and its records are organized into pages in the file.
- Based on the relation's schema and access pattern, the database will determine: (1) type of file used, (2) how pages are organized in the file, (3) how records are organized on each page, (4) and how each record is formatted.

SSNz	Last Name	First Name	Age	Salary
123	Adams	Elmo	31	\$400
443	Grouch	Oscar	32	\$300
244	Oz	Bert	55	\$140
134	Sanders	Ernie	55	\$400



File

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File Organization in DBMS

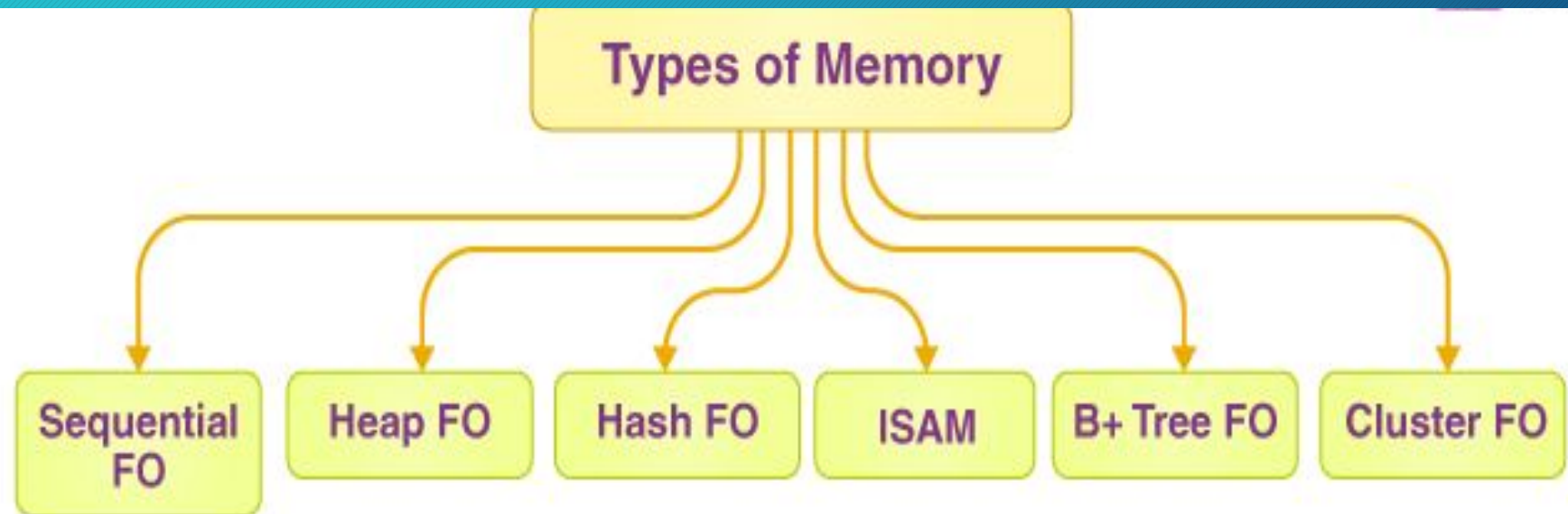
- The logical relationships among the many records that make up the file, particularly in terms of means of identification and access to any given record, are referred to as file organization. Simply put, file organization is the process of storing files in a specific order.
- This method specifies how disc blocks are mapped to file records. The word “file organization” refers to the method by which records are organized into blocks and then placed on a storage media.
- The first method of mapping a database to a file is to employ many files, each containing only one fixed-length entry. Another option is to arrange our files so that we can store data of various lengths. Fixed-length record files are easier to implement than variable-length record files.

Objectives of File Organization in Data Structure

- Here are some ways in which file organization helps:
- It has an ideal record selection, which means records can be selected as quickly as feasible.
- Insert, delete, and update transactions on records should be simple and rapid.
- Duplicate records cannot be created by inserting, updating, or deleting records.
- Records should be stored efficiently to save money on storage.

Types of File Organization

- There are a variety of ways to organize your files. On the basis of access or selection, these methods offer advantages and disadvantages. The programmers choose the finest file organization strategy for their needs when it comes to file organization.

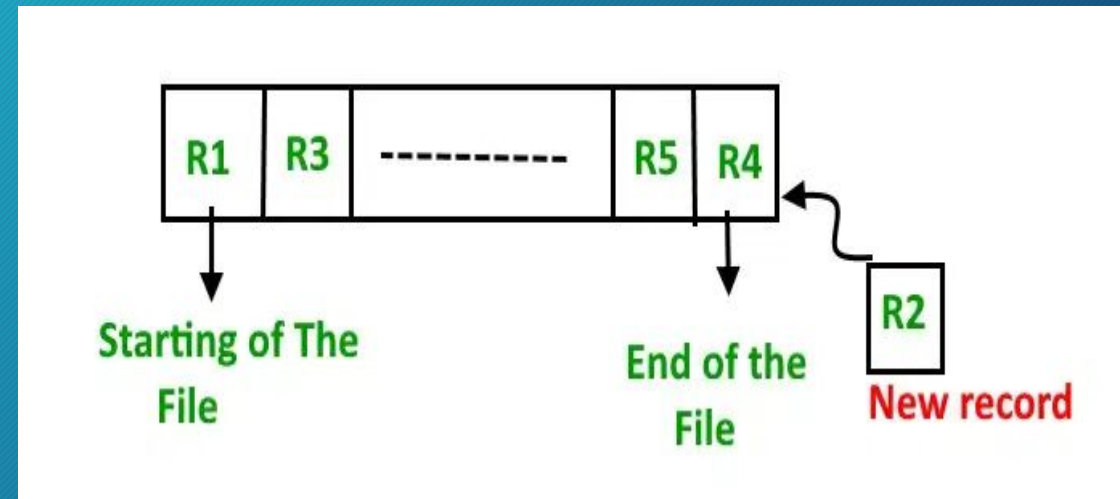


Types of File Organization

- The following are the types of file organization in DBMS:
- Sequential File Organization
- Hash File Organization
- Heap File Organization
- Clustered File Organization
- B+ Tree File Organization
- Indexed Sequential Access Method or ISAM

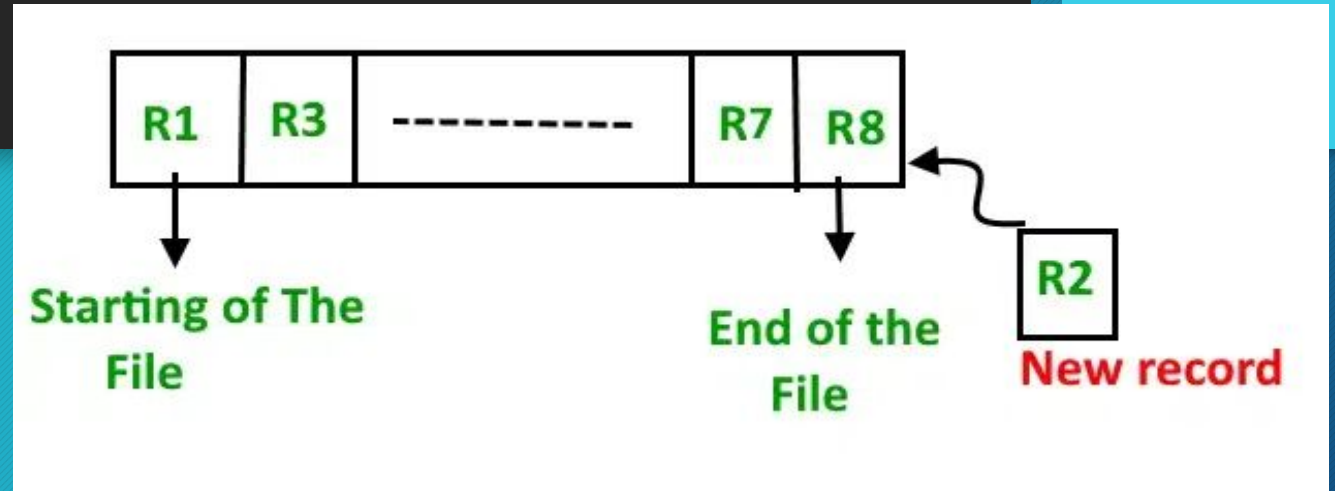
Sequential File Organization

- As the name suggests, this method simply stores the records in files in sequential order, one after another in a series like a sequence of books on a bookshelf. To access these files, we must search through the entire sequence until we reach our desired file in $O(n)$. otherwise, we can use binary search to access in $O(\log n)$.
- Depending on the ordering structure of the records, there are two ways to arrange them sequentially.
- **Pile File Method**
- Records are stored in this method sequentially, one after another, and they are inserted at the end of the file in the same order that we insert them in the table using the SQL query, so it is just an $O(1)$ space complexity operation since the order of the records does not matter.



Sorted File Method

- As the name implies, the file in this method must always be kept in sorted order. In this approach, the file is sorted using a primary key or another reference after each delete, insert, and update operation.



Advantages AND Disadvantages

- **Advantages**

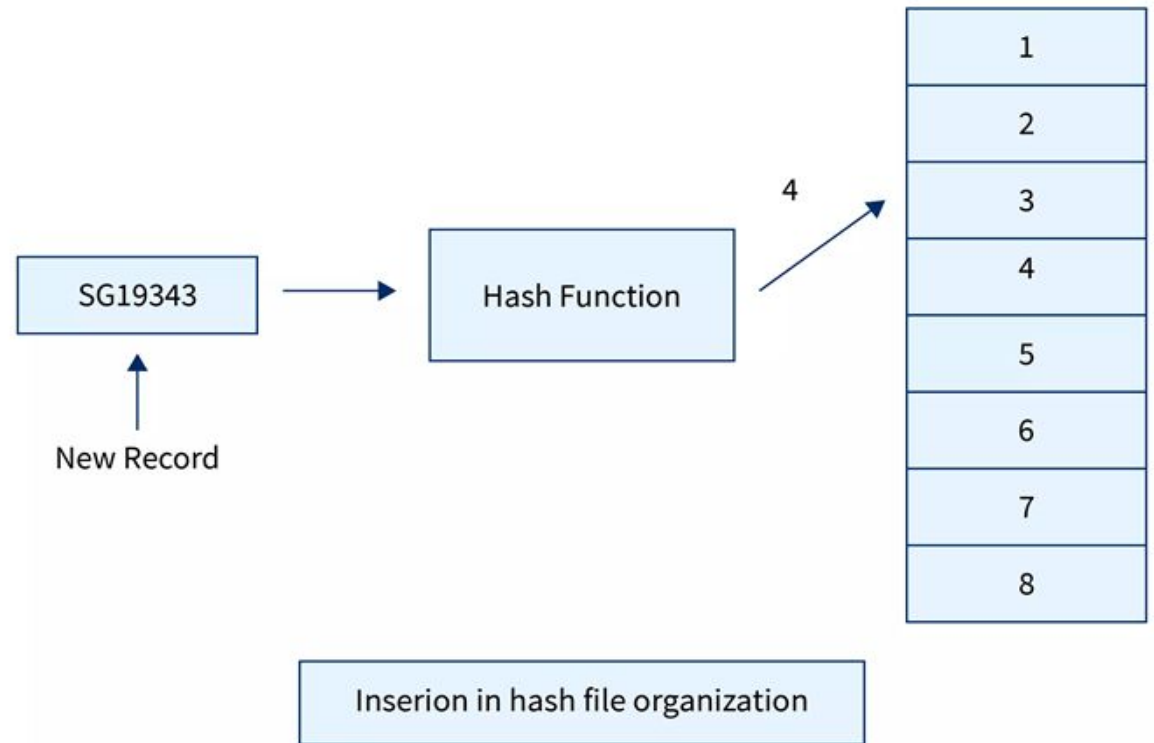
- A fast and effective approach for handling massive amounts of data.
- Plain design
- Files can be easily saved on magnetic tapes, which is a less expensive storage method.

- **Disadvantages**

- Time is wasted because we have to move sequentially and take our time instead of jumping right to the record that is needed.
- The sorted file method wastes time and space sorting records, making it inefficient.

Hash File Organization

- Hashing is used in hash file organization to create the addresses of the memory blocks where actual records are mapped.
- In essence, a hashing function creates the address of those data blocks using the primary key as input, and the memory locations created by these hash functions are referred to as data buckets or data blocks.



Hash File Organization :Advantages & Disadvantages

Advantages

Hash file organization in DBMS uses a hashing function that quickly returns the bucket address of any record, making it a very quick and effective method. It is used in large databases for online banking, e-commerce, and ticketing because it works quickly and effectively.

Due to each record's independence and the ability to access multiple records concurrently, hash file organization in DBMS can handle multiple transactions simultaneously.

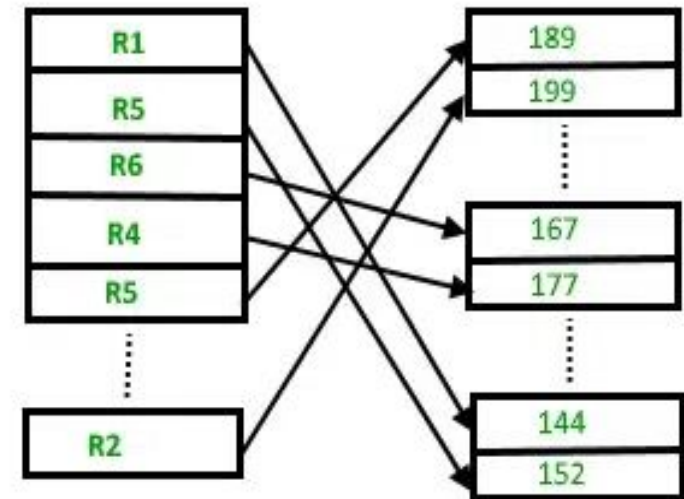
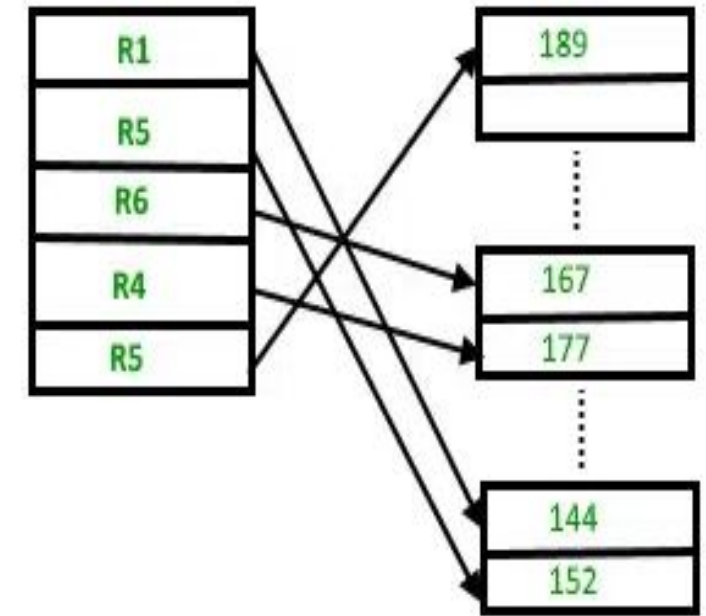
Disadvantages

There is only one column that can be searched, the one that is used to hash bucket addresses. All other columns cannot be searched.

Because the hash is a randomly generated bucket address and lacks order, memory is not utilized

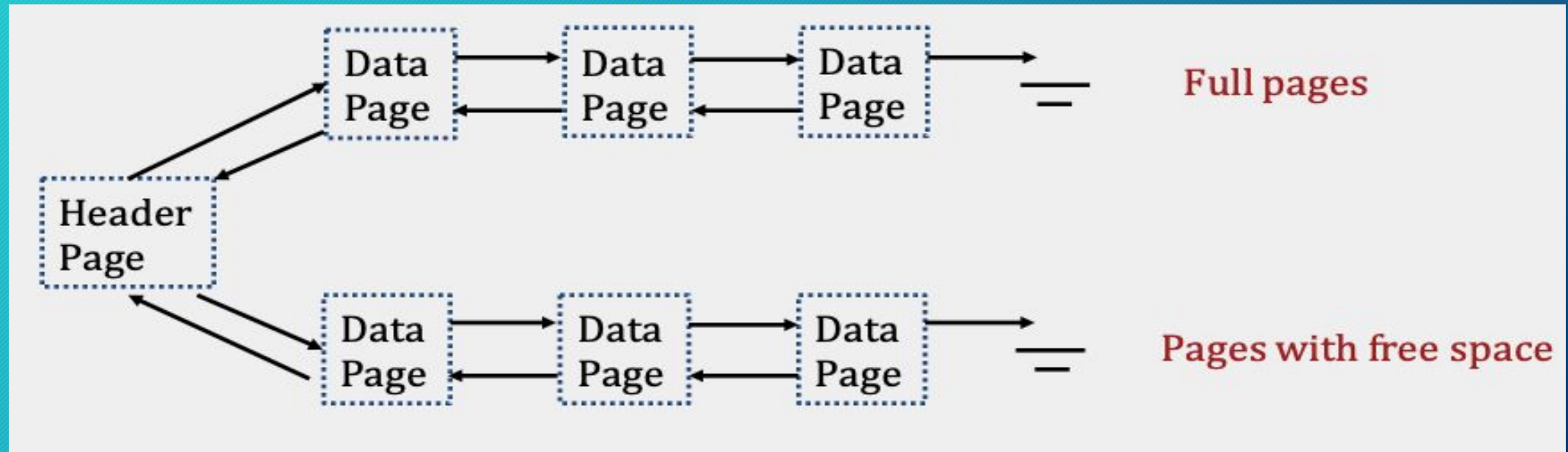
Heap File Organization

- A heap file is a file type with no particular ordering of pages or of the records on pages and has two main implementations. This method doesn't call for any sorting or ordering. The new record is stored in a new block if a data block is full. In this case, the other block does not necessarily have to be the next data block; it can be any block in the memory.
- **Advantages**
 - Fetching and retrieving records is quicker for small databases than sequential record access.
 - This file organization method works best when a significant amount of data must be loaded into the database.
- **Disadvantages**
 - Unused memory blocks are the problem.
 - Larger databases find it ineffective.



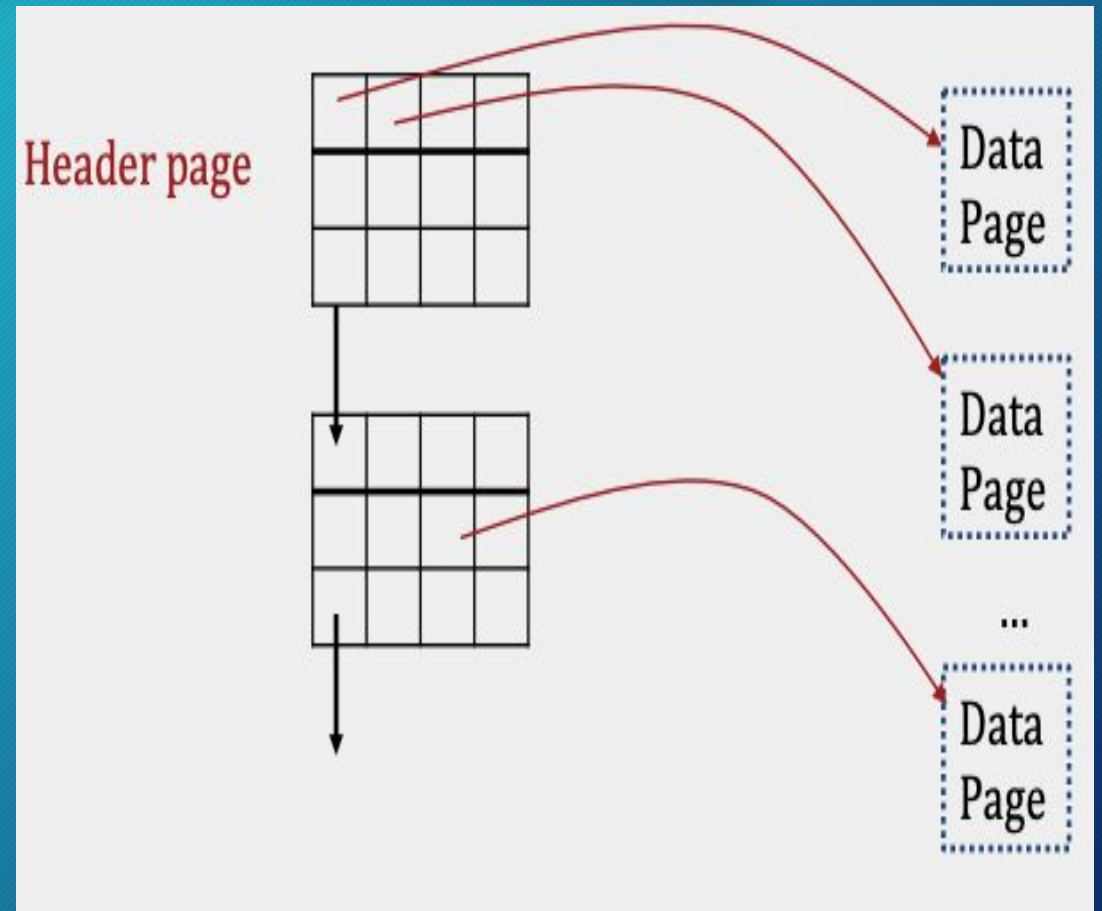
Heap :-Linked List Implementation

- In the linked list implementation, each data page contains records, a free space tracker, and pointers (byte offsets) to the next and previous page. There is one header page that acts as the start of the file and separates the data pages into full pages and free pages.



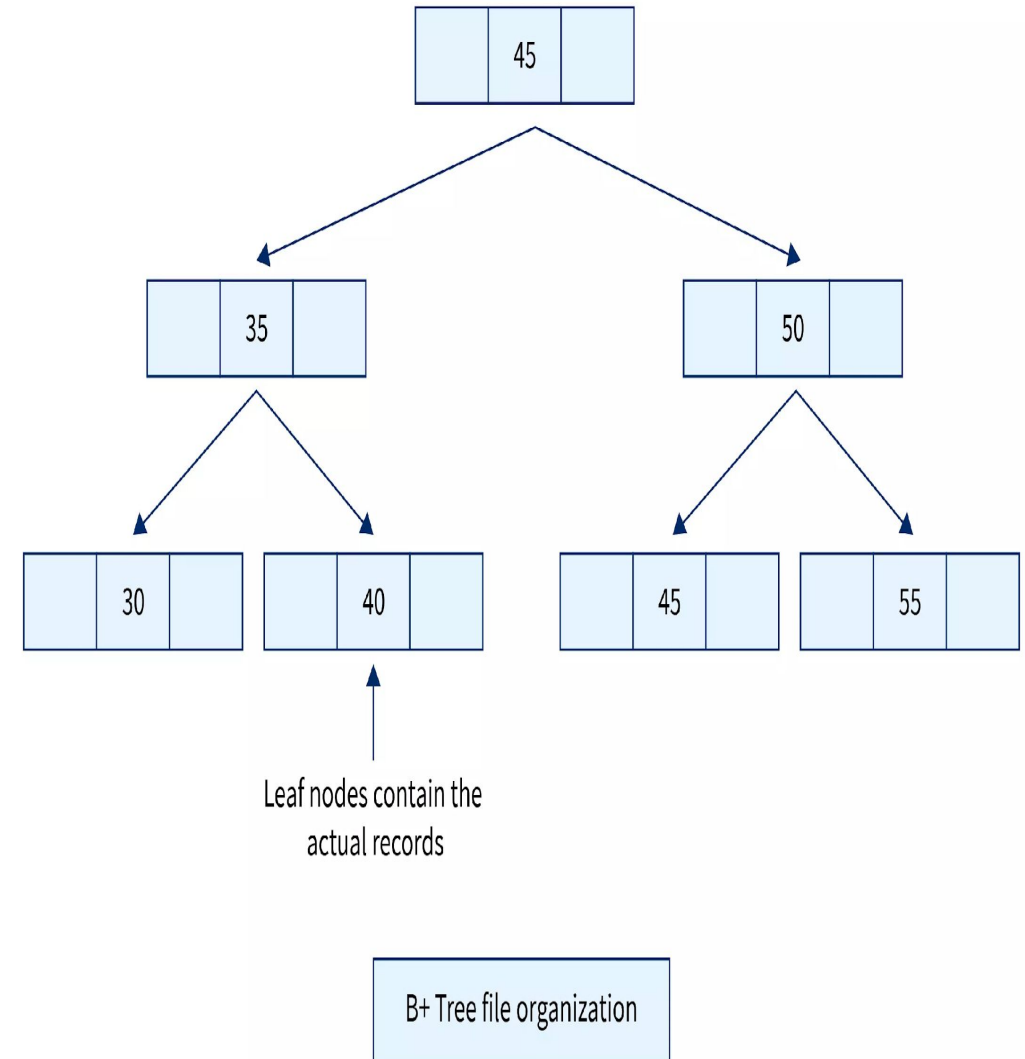
Heap:Page Directory Implementation

- The Page Directory implementation differs from the Linked List implementation by only using a **linked list for header pages**.
- Each header page contains a pointer (byte offset) to the next header page, and its entries contain both a **pointer to a data page** and the **amount of free space left within that data page**.
- Since our header pages' entries store pointers to each data page, the data pages themselves no longer need to store pointers to neighboring pages.



B+ Tree File Organization

- As its name implies, B+ Tree uses a tree-like structure to store records in a file. According to the key indexing principle, the primary key is used to sort the records. A produced index value is associated with the record for each primary key. An index of a record is a record's file address.
- With the exception of having more children than just two, B+ Tree and binary search tree are extremely similar. The leaf nodes are where all the information is kept, while the intermediate nodes serve as pointers to those nodes. A sorted sequential linked list represents the data in leaf nodes at all times. Let us understand its advantages and disadvantages.



advantages and disadvantages of the B+ tree data structure. .

- **Advantages:**

- The records in B+ trees are in the form of singly linked lists to make the searching of data more efficient and quick.
- B+ tree is a balanced tree structure, and hence any operation delete, insert, or update does not affect the overall performance of the tree.
- Tree traversal is easier and relatively fast.
- The size of the tree is dynamic because it can grow or shrink dynamically according to the number of records.

- **Disadvantages:**

- The drawback of B+ trees is that this method is not efficient for static tables.
- Let's understand the next file organization method.

Clustered file organization

- In clustered file organization, two or more records/tables are combined into a single file based on the clustered key or hash clusters, these files contain two or more tables in the same memory block, and all of them are combined using a single clustered key/hash key to a single table.
- **Example:-** Suppose the database of the university where we have two tables, namely the student table, which gives the details of the student, and the course table, which contains the information about the course.

Student		
Student_ID	S_Name	Course_ID
100	Jack	CS101
200	Raman	CS104
300	Victor	CS305
400	Shwetha	CS101
500	Johnny	CS305
600	Himanshu	CS101

Course	
Course_ID	S_Name
CS101	DBMS
CS305	OS
CS104	OOPS

Clustered Key			
Student_ID	S_Name	Course_ID	C_Name
100	Jack	CS101	DBMS
200	Raman	CS104	OOPS
300	Victor	CS305	OS
400	Shwetha	CS101	DBMS
500	Johnny	CS305	OS
600	Himanshu	CS101	DBMS

Let's understand when we need to use clustering in tables.

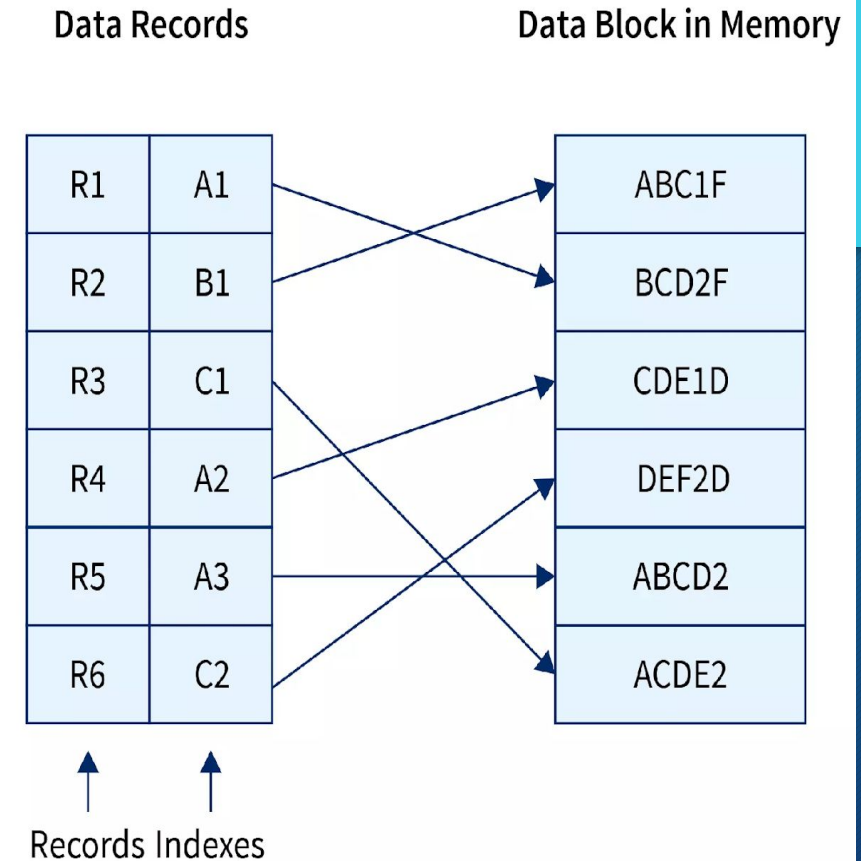
- In the above case, we want to retrieve the students who are enrolled in a particular course, hence we need to join them first and then perform the query for selecting such records, and we join the tables every time we want to retrieve the data so to avoid these computations we combine them into a single table based on a particular clustered index which is Course_id in this case.
- This operation will save time for us as we just need to run the query for the combined table and no longer need to use the join operation.
- Whenever we have a one-to-many relationship between the tables, then we opt for the clustered file organization method in DBMS as in the above example, one course can be opted by many students; hence there is a 1:M relationship between the course and student table.
- Whenever we need to use the join operation very frequently for joining the tables of the database, then we may consider clustering those tables.

Types of Clustered File Organization:

- There are two types of clustered file organization methods in DBMS
- **1) Indexed Clusters:** In this case, the tables are combined on the basis of the clustered key for example, in the above case where student and course tables are on the basis of course_id.
- **2) Hash Clusters:** In this case, the tables are combined on the basis of the hash value of the clustered keys, and we store the results on the basis of the same hash key value.
- Let's understand the Advantages and Disadvantages of the Clustered File Organization in DBMS:
- **Advantages:**
 - The cluster file organization in DBMS is used when there is a join operation for the tables.
 - When there is a 1:M relationship, then this method is efficient.
- **Disadvantages:**
 - Clustered file organization in DBMS is inefficient for the less frequently joined tables and with the tables with one-to-one relationships.
 - Clustered file organization in DBMS is inefficient for large databases.

Indexes Sequential Access Method

- This method is an advanced file organization in DBMS in which for each record in a file, an index value is generated from its primary key, and that index value is mapped with the record, this index contains the address of the record as shown below:
- Let's understand the advantages and disadvantages of this file organization method in DBMS:



Indexes sequential file access

Advantages and disadvantages of (ISAM) file organization method in DBMS:

- **Advantages:**

- Since there is an index corresponding to each record in the table, it is quicker to access any record in the memory, hence ISAM file organization in DBMS can be used for managing large databases.
- Range retrieval and partial retrieval are possible in this method since the index is generated from the key value column we can generate the record addresses of a range of key values, also when a partial key is provided, like student names starting with "RA " can also be searched efficiently.

- **Disadvantages:**

- The main disadvantage is that the ISAM file organization in DBMS takes a lot of space for storing index values; hence when the records increase in number the number of indexes also increases.

Objective of file organization

- File organization in DBMS makes the selection of the records, i.e., querying the database for some records, easy and optimal by giving an ordered arrangement.
- File organization in DBMS makes the database system efficient for the delete, insert and update operations.
- File organization in DBMS also checks for redundancy in the records as a result of the insert, delete, or update operation.
- File organization in DBMS tries to minimize the storage cost by storing them efficiently using the right data structures and techniques.

Conclusion

- At the low level, all the tables that we create using SQL are stored in the form of files in the physical memory.
- The memory is divided into the disk blocks where records are mapped through spanned or unspanned mapping.
- File organization is a method to organize records in a database using different storage techniques like hashing, sequential, B+ tree etc., so as to make the database efficient for the insert, delete, and update operations.
- Depending upon the size of the database and type of records, the frequency of accessing the right method for file organization is chosen to ensure the most efficient database.